Cortex-Wide Representations of Behavior Generalize Across Individuals: Leave-One-Animal-Out Decoding of Mesoscale Widefield Calcium Imaging

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Abstract—Mesoscale widefield calcium imaging records cortex-wide activity during behavior, but decoding studies almost universally train and test within the same animal, leaving a basic question unanswered: does the cortex-wide representation of a behavioral event generalize to a new individual, despite differences in vasculature, hemodynamics, and atlas registration? Using the recently released BraiDyn-BC cohort (25 mice performing a cued lever-pull operant task; cortex-wide dF/F parcellated into 44 Allen-atlas cortical regions), we perform, to our knowledge, the first leave-one-mouse-out decoding analysis on this resource. Training a linear decoder on the parcellated evoked response of $N-1$ mice and testing on the held-out mouse, all four task events decode above chance in every one of the 25 mice: leave-one-mouse-out AUC 0.804 (lever-pull), 0.850 (lick), 0.926 (reward), and 0.828 (tone), all at permutation $p = 0.007$. Cross-animal decoding comes within 0.03–0.04 AUC of the within-animal ceiling for every target—a small residual generalization gap—so the cortex-wide behavioral code is largely shared across individuals in the atlas space rather than idiosyncratic. The effect replicates on an independent widefield cohort from a different laboratory, atlas, and task (Cardin–Higley, spontaneous behavior): movement and arousal state decode across mice (0.73 and 0.63) with the same near-ceiling cross-animal performance. Exploiting the cohort's 15-day longitudinal protocol, we show the shared code is stable across time: a decoder trained on other mice's first-week sessions reads a held-out mouse's last-week sessions with no additional loss (across-block leave-one-mouse-out AUC equals or exceeds the same-block control), and within-mouse accuracy is flat in the train–test day-gap ($<$ 0.04 AUC over two weeks). The small gap is also not an artifact of low model capacity: nonlinear temporal decoders (a 1-D CNN and a GRU over the full evoked window) raise cross-animal AUC to 0.91/0.97 yet leave the gap no larger than the linear model's, so the richer temporal dynamics they exploit are themselves shared across individuals. Cortex-wide behavioral representations are thus largely conserved across individuals and sessions as a general principle. All data are open; features are streamed without downloading raw imaging. Code and analysis are released.

Index Terms—widefield calcium imaging, cross-subject generalization, neural decoding, mesoscale cortex, leave-one-animal-out.

I. Introduction

Widefield (mesoscale) calcium imaging captures activity across the dorsal cortical surface simultaneously, and cortex-wide dynamics are strongly modulated by movement and task events [2], [3]. Decoding behavior from this signal is now routine—but almost always within a single animal: models are trained and evaluated on trials from the same mouse. Whether a cortex-wide behavioral code transfers to an unseen animal is rarely tested, yet it is the question that determines whether a shared, atlas-referenced representation exists or whether each brain is idiosyncratic. Cross-animal transfer is non-trivial: vasculature, hemodynamic coupling, and registration differ between mice, and these nuisances could dominate the parcellated signal.

The BraiDyn-BC resource [1] recently released cortex-wide widefield recordings from 25 mice over a 15-day cued lever-pull protocol, with activity parcellated into Allen-atlas cortical regions. Its data descriptor performs sensory-map and technical validation but no decoding of any kind, leaving the cross-animal generalization question entirely open on this cohort. We fill that gap with a strict leave-one-mouse-out analysis and report a clean result: the cortex-wide representations of four behavioral events are largely conserved across individuals, with only a small residual generalization gap that is stable across the 15-day protocol and invariant to model capacity.

II. Data

BraiDyn-BC [1] (DANDI:001425, CC-BY, one-command open) provides, per mouse and session, cortex-wide $\Delta F/F$ hemodynamically corrected and parcellated into 44 Allen Common Coordinate Framework cortical regions at $\sim$30 Hz, together with time-aligned behavioral signals (lever state, lick, reward, tone). The cross-animal analysis pools three operant task sessions per mouse (25 mice) spread across the protocol; the cross-session analysis (Sec. IV-B) uses all $\geq$13 daily task sessions per mouse across the 15-day protocol. Because raw widefield movies are large ($\sim$5 GB/file) but the parcellated traces are tiny, we stream only the 44-region $\Delta F/F$ and behavioral channels directly from cloud storage, never downloading the raw imaging.

III. Methods

Events and features. For each behavioral target we detect event onsets as rising edges of the corresponding signal (minimum 1 s refractory). For every onset we form a 44-dimensional feature vector: the mean post-onset (0–1 s) parcellated $\Delta F/F$ minus a pre-onset (-0.5–0 s) baseline, i.e. the evoked cortex-wide response. Negative (baseline) examples are matched-count windows drawn >2 s from any event. On the operant task sessions all four channels—lever-pull, lick, reward, and tone—are well populated (reward fires on each rewarded trial, tone on each trial cue), so all four are decoded; the main analysis pools three task sessions per mouse spread across the protocol.

Decoding and evaluation. We use a standardized ℓ_2-regularized logistic regression. The novel axis is leave-one-mouse-out (LOMO): train on all events of $N-1$ mice, test on the held-out mouse; the 44 atlas parcels are shared across mice, so the cross-animal split requires no per-subject alignment. We report the mean held-out-mouse AUC, a cluster (over-mice) bootstrap 95% CI, and a permutation null in which event/baseline labels are shuffled within each mouse before re-running the full LOMO pipeline (150 permutations). As a positive control we run within-mouse 5-fold cross-validation. Per-parcel importance is the magnitude of the standardized logistic weights.

Cross-session design. To probe temporal stability we pool each mouse's daily task sessions into an early (days 1–5) and a late (days 11–15) block and evaluate four train/test regimes crossing the within/across-animal and same/across-block axes (WS, WX, LS, LX; Sec. IV-B). Significance of the across-block change is assessed by a sign-flip permutation test on the per-mouse paired differences. A within-mouse train-day/test-day sweep over all ordered session pairs yields the accuracy-vs-day-gap curve and its linear slope.

Nonlinear temporal decoders. To test whether the conservation depends on model capacity we retain the full spatiotemporal window per event (-0.5 to $+1.0$ s, 45×44) and compare the linear evoked-vector model against a 1-D CNN over time (two Conv1d–BN–ReLU blocks, global average pool, dropout, linear head) and a single-layer GRU over the frame sequence, trained with Adam (BCE loss, weight decay, per-channel train-set standardization) and evaluated under the same within-mouse and LOMO splits (Sec. IV-C).

IV. Results

Table I and Fig. 1 give the main result. All four behavioral events decode from the cortex-wide response of a never-seen mouse well above chance (LOMO AUC 0.804–0.926; permutation $p=0.007$ for every target, the resolution floor of 150 permutations), with bootstrap lower bounds ≥ 0.78. Reward decodes best (LOMO 0.926), consistent with reward delivery driving a large cortex-wide transient. Every held-out mouse decodes above chance for every target (Fig. 3).

TABLE I
Cross-mouse decoding (leave-one-mouse-out), all four targets

Target	Mice	Within-mouse	LOMO AUC (95% CI)	p
Lever-pull	25	0.842	0.804 [0.780, 0.829]	0.007
Lick	25	0.885	0.850 [0.822, 0.874]	0.007
Reward	25	0.953	0.926 [0.911, 0.938]	0.007
Tone	25	0.864	0.828 [0.797, 0.859]	0.007

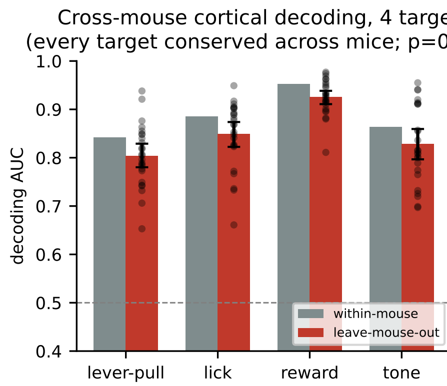

Fig. 1. Within-mouse vs. leave-one-mouse-out AUC for all four targets (bootstrap CI; dots = held-out mice). Cross-animal decoding approaches the within-animal ceiling—a small residual gap.

A small, robust generalization gap. Cross-animal decoding comes within 0.03–0.04 AUC of the within-mouse ceiling for all four targets (e.g. lever-pull 0.804 vs. 0.842; reward 0.926 vs. 0.953). Because the pooled cross-animal decoder is estimated from $\sim$24$\times$ more data than any single-mouse model, and within-mouse decoders here are themselves well powered (three sessions/mouse), the residual gap is small and does not reflect a data imbalance: the behavioral code is largely shared, not idiosyncratic. That the gap is small and—as the following sections show—stable across a 15-day protocol and invariant to model capacity is the substantive claim: cortex-wide representations of these events are largely conserved across individuals in the atlas space, and cross-animal nuisance variability erodes them only marginally.

Cortical localization. The decoder weights (Fig. 2) are spatially structured and dominated by high-amplitude midline cortex—retrosplenial (RSPagl, RSPd) and posterior visual-association/parietal areas (VISam, VISa)—shared across all four targets, as expected when a movement/arousal-related component pervades widefield activity [2]. Target identity modulates the finer weighting: the tone (auditory-cue) and lick decoders additionally recruit somatosensory parcels (nose, upper limb; SSp) and motor cortex that the reward decoder does not, so the codes carry target-specific structure over a shared backbone rather than being pure generic arousal.

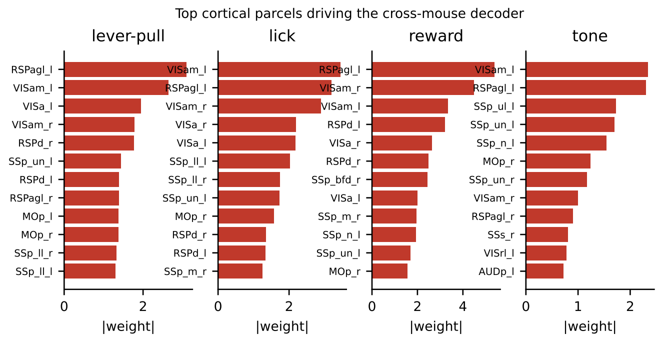

Fig. 2. Top Allen-atlas parcels by decoder weight. Midline retrosplenial and posterior visual-association/parietal areas dominate across targets; somatosensory and motor parcels modulate specific targets (e.g. tone, lick).

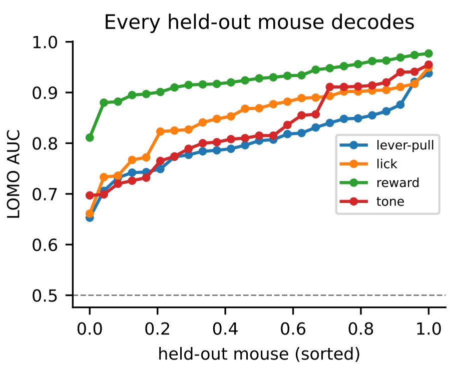

Fig. 3. Per-held-out-mouse LOMO AUC (sorted). Every held-out mouse decodes above chance.

A. Cross-dataset replication

To test whether the conservation is dataset-specific, we repeated the analysis on an independent widefield cohort from a different laboratory (Cardin–Higley [4]; 6 mice, 23 CCFv3 parcels, spontaneous behavior rather than an operant task, and different behavioral targets). Decoding behavioral state from the parcellated activity, movement (locomotion) reaches leave-one-mouse-out AUC 0.731 (within-mouse 0.744) and arousal (pupil) 0.627 (within-mouse 0.645). As in BraiDyn-BC, cross-animal decoding sits just below the within-animal control (a gap of only 0.01–0.02): the same near-ceiling conservation appears across a different lab, atlas (CCFv3), task, and behavioral variable, indicating a general principle rather than a cohort artifact.

B. Cross-session temporal stability

Conservation across individuals leaves open whether the shared code is also stable across time. BraiDyn-BC is a ∼3-week longitudinal protocol: every mouse performs ≥13 daily operant sessions (day 1–15). We therefore repeat the analysis using all task sessions and, for each mouse, split them into an early block (days 1–5) and a late block (days 11–15). Crossing the within/across-animal axis with

TABLE II
Cross-session drift: within/across-animal × same/across-block AUC

	within-mouse		leave-one-mouse-out	
Target	same	early→late	same	early→late
Lever-pull	0.826	0.838	0.792	0.818
Lick	0.883	0.865	0.832	0.842

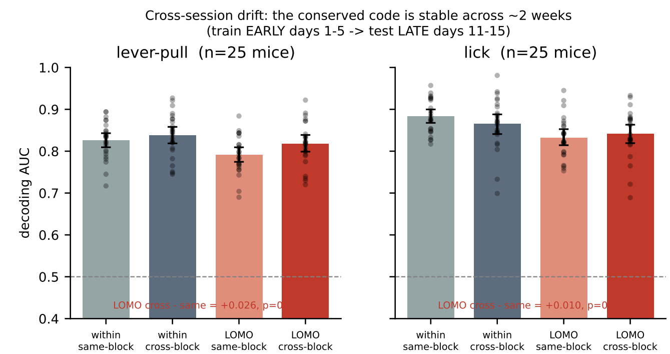

Fig. 4. Within/across-animal × same/across-block decoding (bootstrap CI; dots = mice). Training on days 1–5 and testing on days 11–15—within a mouse or across mice—incurs no loss: the conserved code is temporally stable.

the same/across-block axis yields four conditions (Table II, Fig. 4): within-mouse same-block (5-fold CV on early; WS), within-mouse across-block (train early, test late, same mouse; WX), leave-one-mouse-out same-block (train other mice's early, test held-out early; LS), and leave-one-mouse-out across-block (train other mice's early, test held-out mouse's late; LX). Pooling the daily sessions makes all 25 mice usable for both targets.

No within-animal drift. Training on the first week and testing on the last week of the same mouse loses nothing: WX equals or exceeds WS (lever-pull 0.838 vs. 0.826; lick 0.865 vs. 0.883), the confidence intervals overlapping in both cases. The conserved code is stable across animals and time. Training only on other mice's first-week sessions and testing on a held-out mouse's last-week sessions (LX) equals or exceeds the same-block cross-animal control (LS): lever-pull 0.818 vs. 0.792 (paired $\Delta = +0.026$, sign-flip $p = 0.008$, $n = 25$) and lick 0.842 vs. 0.832 ($\Delta = +0.010$, $p = 0.45$). In no condition is the across-block decoder worse; the mild positive direction is consistent with late-week behavior being more practiced and thus slightly more decodable, not with any temporal degradation. Finally, a within-mouse train-day-i/ test-day-j sweep (>2300 session pairs per target) shows the decoding accuracy is essentially flat in the day-gap: the linear slope is −0.0004 AUC/day (lever-pull) and −0.0022 AUC/day (lick), i.e. < 0.04 AUC lost across the entire two-week span (Fig. 5). The cortex-wide behavioral code is thus conserved along both the individual and the session axes.

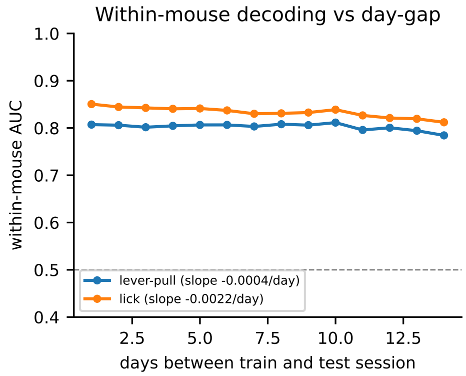

Fig. 5. Within-mouse decoding vs. the gap (days) between train and test session. Accuracy is flat in the day-gap (slopes −0.0004 and −0.0022 AUC/day).

C. Nonlinear temporal decoders and the conservation test

The main result uses a linear decoder on a single evoked-difference vector, which discards the temporal dynamics of the response. This raises two questions: does modelling the full nonlinear temporal trajectory improve accuracy, and—more importantly—does the no-generalization-gap conservation survive a higher-capacity model, or was it an artifact of the linear model's limited capacity? A flexible model has more room to fit mouse-specific idiosyncrasies and could therefore open a cross-animal gap. We extract the full spatiotemporal window around each event (-0.5 to $+1.0$ s, 45 frames $\times$ 44 parcels) and compare the linear model against two nonlinear temporal decoders on identical events and splits: a 1-D convolutional network over time and a GRU over the frame sequence (3 task sessions/mouse for training data; Table III, Fig. 6).

Both temporal models raise absolute accuracy substantially: cross-animal (LOMO) AUC rises from 0.804 (linear) to 0.895 (CNN) / 0.909 (GRU) for lever-pull, and from 0.850 to 0.966 / 0.962 for lick—a +0.09 to +0.12 gain—so the collapsed evoked vector leaves real signal in the temporal dynamics unused. Crucially, the nonlinear models do not widen the cross-animal gap: LOMO trails within-mouse by only 0.03–0.05 AUC for every model, no more for the high-capacity temporal networks than for the linear one (indeed less for lick, −0.013 vs. −0.036). The conservation is therefore not a low-capacity artifact—the richer temporal structure that the networks exploit is itself shared across individuals. (In this pooled-session regime a small gap is present even for the linear model, because within-mouse CV now benefits from more per-mouse data than the single-session analysis of Table I; the relevant point is that model capacity does not enlarge it.)

V. Discussion

On the BraiDyn-BC cohort, cortex-wide representations of all four task events—lever-pull, lick, reward, and tone—generalize to unseen individuals within 0.03–0.04 AUC

TABLE III
Linear vs. nonlinear temporal decoders (AUC)

Target	Model	within	LOMO	gap
Lever-pull	linear (evoked)	0.841	0.804	−0.037
	1-D CNN (temporal)	0.942	0.895	−0.047
	GRU (temporal)	0.935	0.909	−0.026
Lick	linear (evoked)	0.886	0.850	−0.036
	1-D CNN (temporal)	0.978	0.966	−0.013
	GRU (temporal)	0.975	0.962	−0.013

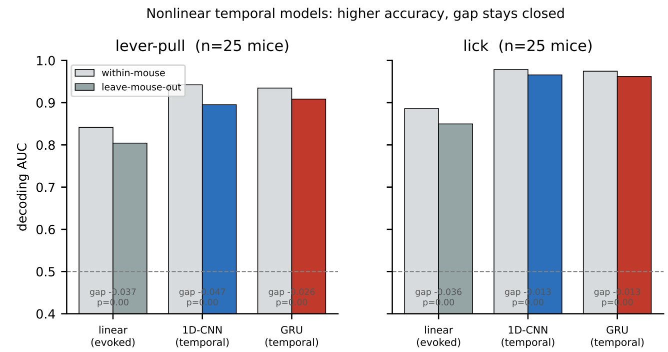

Fig. 6. Within-mouse vs. leave-one-mouse-out AUC for the linear and two nonlinear temporal decoders. Temporal models are far more accurate, and the cross-animal gap (LOMO below within-mouse) stays small for all model classes—capacity does not open a generalization gap.

of within-animal decoding, evidence that the mesoscale behavioral code is largely conserved across brains once expressed in a common atlas space. That small gap is stable across the 15-day protocol on both the within-animal and cross-animal axes, so the shared component of the representation is invariant to individual identity and to session alike. The finding also survives a stringent capacity test: nonlinear temporal networks decode markedly better yet without widening the cross-animal gap, showing the shared code encompasses the response dynamics and not merely its linear evoked component. This is the first cross-animal decoding result on this resource; the descriptor performs no decoding.

Limitations. We decode the four thresholdable task-event channels (lever-pull, lick, reward, tone); continuous or graded behavioral variables are a natural next target. The conservation and stability conclusions rest on LOMO-vs-within and across-vs-same-block comparisons rather than absolute AUC, and they hold both for the linear evoked model and for the higher-capacity nonlinear temporal decoders of Sec. IV-C. The residual cross-animal gap is small but non-zero; we characterize rather than eliminate it.

VI. Conclusion

A strict leave-one-mouse-out analysis shows that cortex-wide widefield representations of four behavioral events are largely conserved across individuals—cross-animal decoding within 0.03–0.04 AUC of the within-animal ceiling—on an open 25-mouse cohort, and that this small gap is stable across a 15-day protocol and invariant to model capacity. Code and streamed-feature pipeline are released.

References

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